

SAN chemicals

2025-12-14

```
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.5.1
```

```
library(haven)
```

```
## Warning: package 'haven' was built under R version 4.5.1
```

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.5.1
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.5.1
```

```
##  
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':  
##  
##   recode
```

```
library(survey)
```

```
## Warning: package 'survey' was built under R version 4.5.2
```

```
## Loading required package: grid
```

```
## Loading required package: Matrix
```

```
## Loading required package: survival
```

```
##  
## Attaching package: 'survey'
```

```
## The following object is masked from 'package:graphics':  
##  
##      dotchart
```

```
library(srvyr)
```

```
## Warning: package 'srvyr' was built under R version 4.5.2
```

```
##  
## Attaching package: 'srvyr'
```

```
## The following object is masked from 'package:stats':  
##  
##      filter
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.1
```

```
library(tidyr)
```

```
##  
## Attaching package: 'tidyr'
```

```
## The following objects are masked from 'package:Matrix':  
##  
##      expand, pack, unpack
```

Read datasets

```
#Nacist data a z nekterych vybrat jen nektere promenne

#Demographics, RIDAGEYR - vek, RIDEXPRG - pregnant, DMDEDUC2 - education level, RIAGENDR - gender, INDFMPIR - ratio of family income to poverty
#RIDSTATR - both interview and MEC examination (2 - both, 1 - only interview)
demo_data = read_xpt("./data/DEMO_H.xpt") |> select(SEQN,RIDAGEYR,RIDEXPRG,DMDEDUC2,RIAGENDR,INDFMPIR,WTMEC2YR,SDMVPSU,SDMVSTRA)
#Reproductive health data, RHQ031 - had regular periods in past 12 months, RHQ074 - tried for a yer to become pregnant, RHQ076 - seen DR unable to become pregnant, RHQ131 - ever been pregnant,
#RHQ160 - how many times have been pregnant, RHQ171 - live birth deliveries, RHD280 - had a hysterectomy, RHQ078 - pelvic infection,RHD180 - age at first livebirth
rhq_data = read_xpt("./data/RHQ_H.xpt") |> select(SEQN,RHQ031,RHQ074,RHQ076, RHQ131,RHQ160, RHQ171, RHD280, RHQ078, RHD180)
#SEX steroid hormone, LBXTST - testosterone (ng/dL), LBXEXT - estradiol(pg/mL), LBXSHGB (nmol/L), promene koncici na LC (0 - above detection limit, 1 below detection limit)
tst_data = read_xpt("./data/TST_H.xpt")
#Phthalates and plasticizers metabolites
ssphte_data = read_xpt("./data/SSPHTTE_H.xpt")

#Personal care and consumer product chemicals and metabolites
ephpp_data = read_xpt("./data/EPHPP_H.xpt")

#phthalates and plasticizers metabolites
phthte_data = read_xpt("./data/PTHTE_H.xpt")

#alcohol use, ALQ120Q - How often drink alcohol over past 12 months
alq_data = read_xpt("./data/ALQ_H.xpt") |> select(SEQN,ALQ120Q)
#smoking cigarette use, SMQ040 - do you now smoke cigarettes, SMD650 avg cigarettes a day last month
smq_data = read_xpt("./data/SMQ_H.xpt") |> select(SEQN,SMQ040,SMD650)
#medical condition
#mcq_data = read_xpt("./data/MCQ_H.xpt") |> select(SEQN,)
#body measures, BMXBMI - body mass index
bmx_data = read_xpt("./data/BMX_H.xpt") |> select(SEQN,BMXBMI)
```

Merging datasets

```
#inner join keep only rows where there is a SEQN match in both datasets
merged_data = inner_join(demo_data, rhq_data, by = "SEQN")
merged_data = inner_join(merged_data, tst_data, by = "SEQN")
merged_data = inner_join(merged_data, ssphte_data, by = "SEQN")
merged_data = inner_join(merged_data, ephpp_data, by = "SEQN")
merged_data = inner_join(merged_data, alq_data, by = "SEQN")
merged_data = inner_join(merged_data, smq_data, by = "SEQN")
merged_data = inner_join(merged_data, bmx_data, by = "SEQN")
merged_data = inner_join(merged_data, phthte_data, by = "SEQN")
nrow(merged_data)
```

```
## [1] 917
```

```
#jedna promena se vyskytuje ve vice datasetech
merged_data <- merged_data[, !grepl("\\.y$", names(merged_data))]
names(merged_data)[names(merged_data) == "URXUCR.x"] <- "URXUCR"
names(merged_data)[names(merged_data) == "WTSB2YR.x"] <- "WTSB2YR"

var_labels <- sapply(merged_data, function(x) attr(x, "label"))
labels_df <- data.frame(
  variable = names(var_labels),
  label = unname(var_labels)
)
print(labels_df)
```

##	variable	label
## 1	SEQN	Respondent sequence number
## 2	RIDAGEYR	Age in years at screening
## 3	RIDEXPRG	Pregnancy status at exam
## 4	DMDEDUC2	Education level - Adults 20+
## 5	RIAGENDR	Gender
## 6	INDFMPIR	Ratio of family income to poverty
## 7	WTMEC2YR	Full sample 2 year MEC exam weight
## 8	SDMVPSU	Masked variance pseudo-PSU
## 9	SDMVSTRA	Masked variance pseudo-stratum
## 10	RHQ031	Had regular periods in past 12 months
## 11	RHQ074	Tried for a year to become pregnant?
## 12	RHQ076	Seen a DR b/c unable to become pregnant?
## 13	RHQ131	Ever been pregnant?
## 14	RHQ160	How many times have been pregnant?
## 15	RHQ171	How many deliveries live birth result?
## 16	RHD280	Had a hysterectomy?
## 17	RHQ078	Ever treated for a pelvic infection/PID?
## 18	RHD180	Age at first live birth
## 19	LBXTST	Testosterone, total (ng/dL)
## 20	LBDTSTLC	Testosterone comment code
## 21	LBXEST	Estradiol (pg/mL)
## 22	LBDESTLC	Estradiol Comment Code
## 23	LBXSHBG	SHBG (nmol/L)
## 24	LBDSHGLC	SHBG Comment Code
## 25	WTSSBH2Y	Surplus specimen B 13-14 2 year weights
## 26	URXUCR	Creatinine, urine (mg/dL)
## 27	SSURMHBP	Mono-3-hydroxy-n-butyl phthalate (ng/mL)
## 28	SDUMHBPL	Mono-3-hydroxy-n-butyl phthalate cmt
## 29	SSURHIBP	Mono-2-hydroxy-iso-butyl phthlte (ng/mL)
## 30	SDUHIBPL	Mono-2-hydroxy-iso-butyl phthlte cmt
## 31	WTSB2YR	Subsample B weights
## 32	URXBP3	Urinary Benzophenone-3 (ng/mL)
## 33	URDBP3LC	Urinary Benzophenone-3 comment
## 34	URXBPH	Urinary Bisphenol A (ng/mL)
## 35	URDBPHLC	Urinary Bisphenol A comment
## 36	URXBPF	Urinary Bisphenol F (ug/L)
## 37	URDBPFLC	Urinary Bisphenol F comment
## 38	URXBPS	Urinary Bisphenol S (ug/L)
## 39	URDBPSLC	Urinary Bisphenol S comment
## 40	URXTLC	Urinary Triclocarban (ng/mL)
## 41	URDTLCLC	Urinary Triclocarban comment
## 42	URXTRS	Urinary Triclosan (ng/mL)
## 43	URDTRSLC	Urinary Triclosan comment
## 44	URXBUP	Butyl paraben (ng/ml)
## 45	URDBUPLC	Butyl paraben comment
## 46	URXEPB	Ethyl paraben (ng/ml)
## 47	URDEPBLC	Ethyl paraben comment
## 48	URXMPB	Methyl paraben (ng/ml)
## 49	URDMPBLC	Methyl paraben comment
## 50	URXPPB	Propyl paraben (ng/ml)
## 51	URDPPBLC	Propyl paraben comment
## 52	URX14D	2,5-dichlorophenol (ug/L)
## 53	URD14DLC	2,5-dichlorophenol comment
## 54	URXDCB	2,4-dichlorophenol (ug/L)

```

## 55 URDDCBLC          2,4-dichlorophenol comment
## 56 ALQ120Q How often drink alcohol over past 12 mos
## 57 SMQ040           Do you now smoke cigarettes
## 58 SMD650 Avg # cigarettes/day during past 30 days
## 59 BMXBMI           Body Mass Index (kg/m**2)
## 60 URXUCR           Creatinine, urine (mg/dL)
## 61 URXCNP           Mono(carboxynonyl) Phthalate (ng/mL)
## 62 URDCNPLC         Mono(carboxynonyl) Phthalate cmt code
## 63 URXCOP           Mono(carboxyoctyl) Phthalate (ng/mL)
## 64 URDCOPLC         Mono(carboxyoctyl) phthalate cmt code
## 65 URXECp           MECP phthalate (ng/mL)
## 66 URDECPLC         MECP phthalate comment code
## 67 URXMBP           Mono-n-butyl phthalate (ng/mL)
## 68 URDMBPLC         Mono-n-butyl phthalate comment code
## 69 URXMC1 Mono-(3-carboxypropyl) phthalate (ng/mL)
## 70 URDMC1LC         Mono-(3-carboxypropyl) phthalate cmt
## 71 URXMEP           Mono-ethyl phthalate (ng/mL)
## 72 URDMEPLC         Mono-ethyl phthalate comment code
## 73 URXMHh           MEHP phthalate (ng/mL)
## 74 URDMHhLC         MEHP phthalate comment code
## 75 URXMHNC          MHNCH (ng/mL)
## 76 URDMCHLC         MHNCH comment code
## 77 URXMHP           Mono-(2-ethyl)-hexyl phthalate (ng/mL)
## 78 URDMHPLC         Mono-(2-ethyl)-hexyl phthalate cmt code
## 79 URXMIB           Mono-isobutyl phthalate (ng/mL)
## 80 URDMIBLC         Mono-isobutyl phthalate comment code
## 81 URXMNP           Mono-isononyl phthalate (ng/mL)
## 82 URDMNPLC         Mono-isononyl phthalate comment code
## 83 URXMOH           MEOH phthalate (ng/mL)
## 84 URDMOHLC         MEOH phthalate comment code
## 85 URXMZP           Mono-benzyl phthalate (ng/mL)
## 86 URDMZPLC         Mono-benzyl phthalate comment code

```

Remove, based on exclusion, inclusion criteria from the plan

```

#age between 18 and 50 and female only, not pregnant
clean_data <- merged_data[merged_data$RIDAGEYR >= 18 & merged_data$RIDAGEYR <= 50 & merged_data$RIAGENDR == 2 & merged_data$RIDEXPRG == 2,]
nrow(clean_data)

```

```
## [1] 460
```

```
#extreme urinary creatine values
```

```
clean_data <- clean_data[clean_data$URXUCR<300,]
```

```
#missing key covariates -> R si NaN values odstraní automaticky pri vytvareni modelu ?
```

```
#urinary chemicals measurement and MEC examination
```

```
clean_data <- clean_data[clean_data$WTSSBH2Y>0 & clean_data$WTSB2YR>0 & clean_data$WTMEC2YR>0,]
```

```
nrow(clean_data)
```

```
## [1] 448
```

```
#at least one detectable urinary chemical measurement (ze vseh chemikalii v EPHPP a SSPHTE)
```

```
all_vars <- union(union(names(ssphte_data), names(ephpp_data)),names(phthte_data))
```

```
exclude_vars <- c( "SEQN","WTSSBH2Y","WTSB2YR" )
```

```
edc_vars <- setdiff(all_vars, exclude_vars)
```

```
length(edc_vars)
```

```
## [1] 55
```

```
data_1 <- clean_data[
```

```
  rowSums(!is.na(clean_data[, edc_vars])) >= 1, #alespon jedna hodnota ze vseh merenych chemikalii neni Nan (missing)
```

```
]
```

```
# at least one detectable hormone measurement (not missing - nan)
```

```
horm_vars <- c("LBXSHBG","LBXEST","LBXTST")
```

```
data_1 <- data_1[
```

```
  rowSums(!is.na(data_1[, horm_vars])) >= 1, #alespon jedna hodnota ze vseh hormonu neni nan
```

```
]
```

```
nrow(data_1)
```

```
## [1] 324
```

Survey design object -> to work with the weights

zdroj:<https://wwwn.cdc.gov/nchs/nhanes/tutorials/weighting.aspx>

(<https://wwwn.cdc.gov/nchs/nhanes/tutorials/weighting.aspx>) Weights are created in NHANES to account for the complex survey design (including oversampling), survey non-response and post-stratification adjustment to match total population counts in the USA. A sample weight is assigned to each sample person. It is a measure of the number of people in the population represented by that sample person. NHANES: You must use the weight of the smallest subpopulation that includes all the variables you want to include in your analysis. Z tech mensich souboru obsahují speciální vahy jenom ty chemikálie (EPHPP a SSPHTE a PHTHTE). EPHPP a PHTHTE mají stejné (WTSB2YR) a SSPHTE má jiné (WTSSBH2Y). Nevím která z nich by se měla použít, ale nejmenší je dataset SSPHTE. v ukázkách pak na všechny úpravy používají už jenom survey design objekt na práci s weighted NHANES dataset -> vytvořit z celých dat a znovu stejné profilovat

```

data_w <- svydesign(id = ~SDMVPSU, strata= ~SDMVSTRA, weight=~WTSB2YR, nest=TRUE, data=merged_data)

obj_cleaned <- data_w |>
  subset(
    RIDAGEYR >= 18 &
    RIDAGEYR <= 50 &
    RIAGENDR == 2 &
    RIDEXPRG == 2 &
    URXUCR < 300 &
    WTSSBH2Y > 0 &
    WTSB2YR > 0 &
    WTMEC2YR > 0 &
    rowSums(!is.na(as.data.frame(data_w$variables[, edc_vars]))) >= 1 &
    rowSums(!is.na(as.data.frame(data_w$variables[, horm_vars]))) >= 1
  )
nrow(obj_cleaned$variables)

```

```
## [1] 324
```

```
summary(obj_cleaned)
```

```

## Stratified 1 - level Cluster Sampling design (with replacement)
## With (30) clusters.
## subset(data_w, RIDAGEYR >= 18 & RIDAGEYR <= 50 & RIAGENDR ==
##      2 & RIDEXPRG == 2 & URXUCR < 300 & WTSSBH2Y > 0 & WTSB2YR >
##      0 & WTMEC2YR > 0 & rowSums(!is.na(as.data.frame(data_w$variables[,
##      edc_vars]))) >= 1 & rowSums(!is.na(as.data.frame(data_w$variables[,
##      horm_vars)))) >= 1)
## Probabilities:
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 2.515e-06 6.348e-06 1.011e-05 1.047e-05 1.392e-05 3.033e-05
## Stratum Sizes:
##           104 105 106 107 108 109 110 111 112 113 114 115 116 117 118
## obs           13  18  20  26  20  21  15  26  29  22  27  23  20  17  27
## design.PSU    2   2   2   2   2   2   2   2   2   2   2   2   2   2   2
## actual.PSU    2   2   2   2   2   2   2   2   2   2   2   2   2   2   2
## Data variables:
## [1] "SEQN"      "RIDAGEYR" "RIDEXPRG" "DMDEDUC2" "RIAGENDR" "INDFMPIR"
## [7] "WTMEC2YR" "SDMVPSU"  "SDMVSTRA" "RHQ031"   "RHQ074"   "RHQ076"
## [13] "RHQ131"   "RHQ160"   "RHQ171"   "RHD280"   "RHQ078"   "RHD180"
## [19] "LBXTST"   "LBDTSTLC" "LBXEST"   "LBDESTLC" "LBXSHBG"  "LBDSHGLC"
## [25] "WTSSBH2Y" "URXUCR"   "SSURMHBP" "SDUMHBPL" "SSURHIBP" "SDUHIBPL"
## [31] "WTSB2YR"  "URXBP3"   "URDBP3LC" "URXBPH"   "URDBPHLC" "URXBPF"
## [37] "URDBPFLC" "URXBPS"   "URDBPSLC" "URXTLC"   "URDTLCLC" "URXTRS"
## [43] "URDTRSLC" "URXBUP"   "URDBUPLC" "URXEPB"   "URDEPBLC" "URXMPB"
## [49] "URDMPBLC" "URXPPB"   "URDPPBLC" "URX14D"   "URD14DLC" "URXD CB"
## [55] "URDDCBLC" "ALQ120Q"  "SMQ040"   "SMD650"   "BMXBMI"   "URXUCR"
## [61] "URXCNP"   "URDCNPLC" "URXCOP"   "URDCOPLC" "URXECP"   "URDECPLC"
## [67] "URXMBP"   "URDMBPLC" "URXMC1"   "URDMC1LC" "URXMEP"   "URDMEPLC"
## [73] "URXMHH"   "URDMHHL"  "URXMHNC"  "URDMCHLC" "URXMHP"   "URDMHPLC"
## [79] "URXMIB"   "URDMIBLC" "URXMNP"   "URDMNPLC" "URXMOH"   "URDMOHL"
## [85] "URXMZP"   "URDMZPLC"

```

#vytvoreny survey objekt porad obsahuje sloupce s vahama z vetsich datasetu -> pri vytvareni modelu upravit a nebrat je v uvahu

Variables statistics

#age

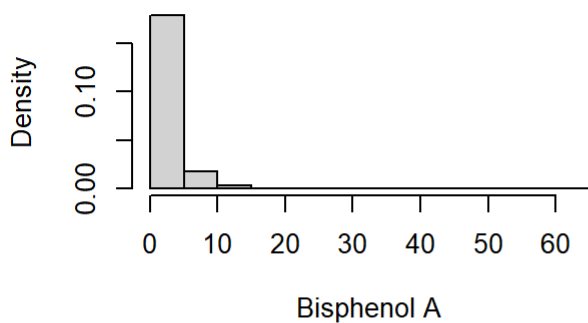
```
svymean(~RIDAGEYR,obj_cleaned,na.rm = TRUE)
```

```
##          mean      SE
## RIDAGEYR 32.326 0.7096
```

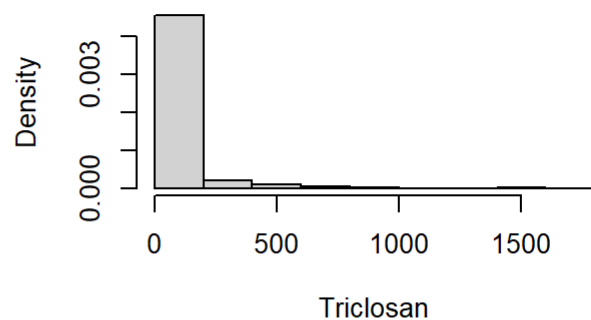
#visualize

```
par(mfrow = c(2, 2))
svyhist(~URXBPH, design=obj_cleaned, main="Bisphenol A concentration",xlab="Bisphenol A")
svyhist(~URXTRS, design=obj_cleaned, main="Triclosan concentration",xlab="Triclosan")
svyhist(~URXMHM, design=obj_cleaned, main="MEHP concentration",xlab="MEHP")
svyhist(~URXMPB, design=obj_cleaned, main="MPB concentration",xlab="MPB")
```

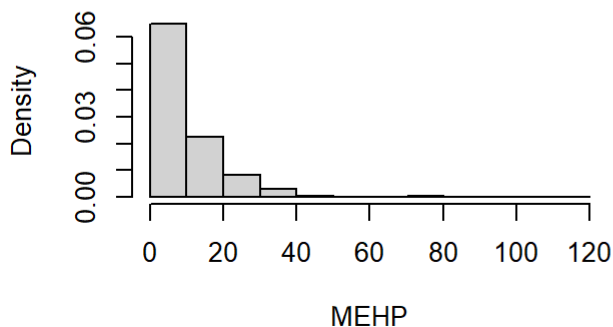
Bisphenol A concentration



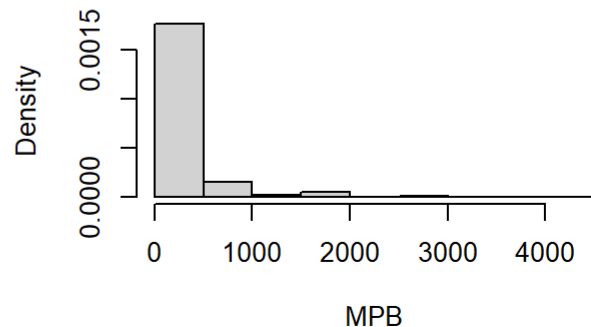
Triclosan concentration



MEHP concentration



MPB concentration



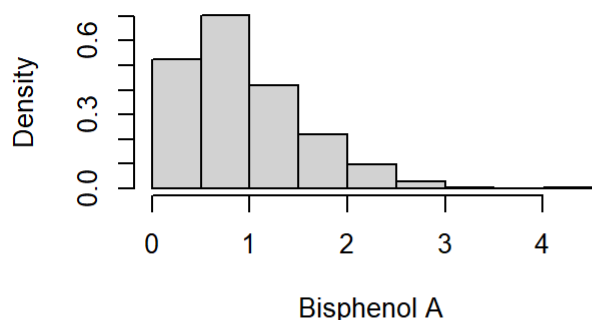
```
par(mfrow = c(1, 1))
```

```

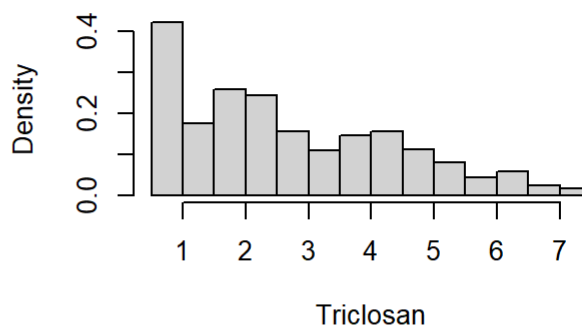
par(mfrow = c(2, 2))
svyhist(~log(URXBPH + 1), design=obj_cleaned, main="Bisphenol A concentration, log transformed",xlab="Bisphenol A")
svyhist(~log(URXTRS + 1), design=obj_cleaned, main="Triclosan concentration, log transformed",xlab="Triclosan")
svyhist(~log(URXMHH + 1), design=obj_cleaned, main="MEHP concentration, log transformed",xlab="MEHP")
svyhist(~log(URXMPB + 1), design=obj_cleaned, main="MPB concentration, log transformed",xlab="MPB")

```

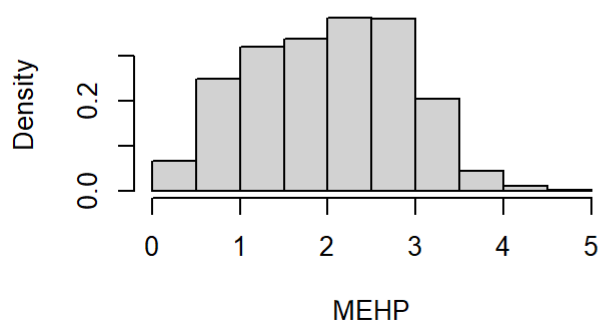
Bisphenol A concentration, log transformed



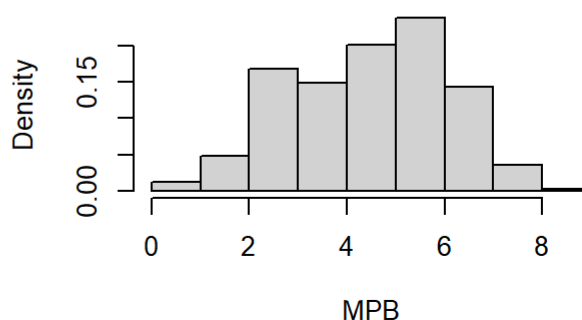
Triclosan concentration, log transformed



MEHP concentration, log transformed



MPB concentration, log transformed



```

par(mfrow = c(1, 1))

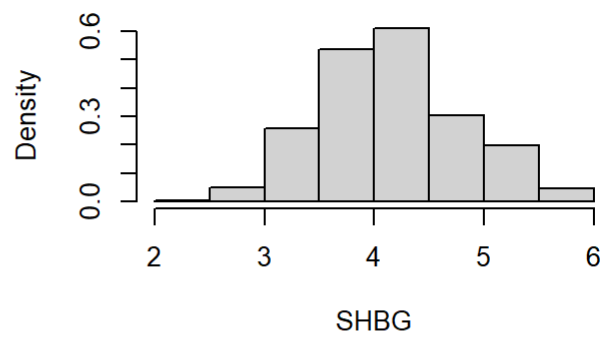
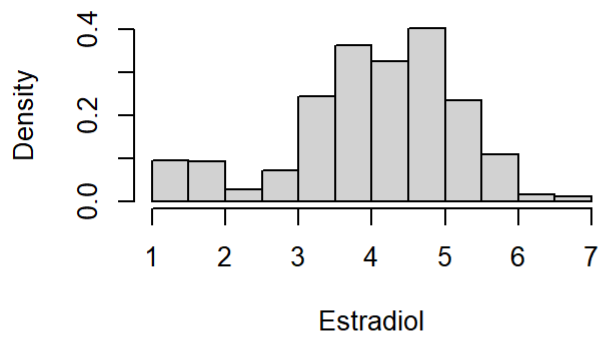
```

```

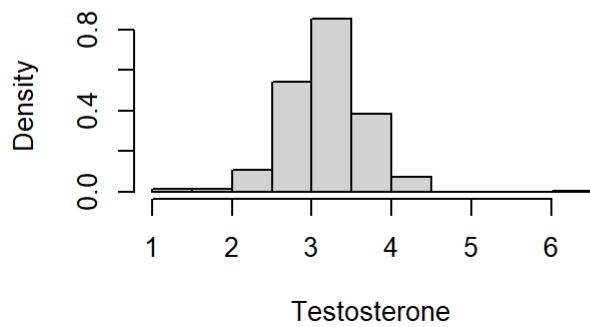
par(mfrow = c(2, 2))
svyhist(~log(LBXEST + 1), design=obj_cleaned, main="Estradiol hormone concentration, log transformed",xlab="Estradiol")
svyhist(~log(LBXSHBG + 1), design=obj_cleaned, main="SHBG hormone concentration, log transformed",xlab="SHBG")
svyhist(~log(LBXTST + 1), design=obj_cleaned, main="Testosterone hormone concentration, log transformed",xlab="Testosterone")
par(mfrow = c(1, 1))

```

estradione hormone concentration, log transformed SHBG hormone concentration, log transformed



testosterone hormone concentration, log transformed



PCA

V clanku (Association between co-exposure to phenols and phthalates mixture and infertility risk in women):
 Use PCA to deal with collinearity of chemicals -> get fewer independent components. Factors with eigen values higher than 1 were determined as the main components. Z 54 chemikalii se to zmensilo na 15 main components.

```

#pridat dalsi sloupce do datasetu s logtransformovanymi chemikalii, edc_vars obsahuje jmena
sloupce vsech chemikalii v datasetu
obj_cleaned$variables[, paste0("log_", edc_vars)] <- log(obj_cleaned$variables[, edc_vars] +
1)
log_edc_vars <- paste0("log_", edc_vars) #log_edc_vars -> jmena vsech log transformovanych ch
emikalii

#pridat i log transformovane hormony do budoucna
obj_cleaned$variables$log_LBXEST <- log(obj_cleaned$variables$LBXEST + 1)
obj_cleaned$variables$log_LBXSHBG <- log(obj_cleaned$variables$LBXSHBG + 1)
obj_cleaned$variables$log_LBXTST <- log(obj_cleaned$variables$LBXTST + 1)

#potreba odstranit tuto chemikalii, jinak error: cannot rescale constant/zero column to unit
variance, je to comment code (value 0 - at or above detection limit)
del <- c( "log_URDMEPLC" )
log_edc_vars <- setdiff(log_edc_vars,del)

pca_w <- svyprcomp(
  ~ as.matrix(obj_cleaned$variables[, log_edc_vars]),
  design = obj_cleaned,
  center = TRUE,
  scale = TRUE
)

summary(pca_w)

```

```
## Importance of components:
##          PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation  3.6704 2.21698 1.66427 1.60227 1.46470 1.3630 1.2835
## Proportion of Variance 0.2495 0.09102 0.05129 0.04754 0.03973 0.0344 0.0305
## Cumulative Proportion 0.2495 0.34050 0.39179 0.43933 0.47906 0.5135 0.5440
##          PC8      PC9      PC10     PC11     PC12     PC13     PC14
## Standard deviation  1.22780 1.17369 1.16733 1.11589 1.09123 1.07147 1.0236
## Proportion of Variance 0.02792 0.02551 0.02523 0.02306 0.02205 0.02126 0.0194
## Cumulative Proportion 0.57189 0.59740 0.62263 0.64569 0.66774 0.68900 0.7084
##          PC15     PC16     PC17     PC18     PC19     PC20     PC21
## Standard deviation  1.00989 0.97708 0.97394 0.95557 0.94227 0.88011 0.85459
## Proportion of Variance 0.01889 0.01768 0.01757 0.01691 0.01644 0.01434 0.01352
## Cumulative Proportion 0.72729 0.74497 0.76254 0.77945 0.79589 0.81023 0.82376
##          PC22     PC23     PC24     PC25     PC26     PC27     PC28
## Standard deviation  0.79780 0.77014 0.74809 0.73141 0.72127 0.7048 0.68712
## Proportion of Variance 0.01179 0.01098 0.01036 0.00991 0.00963 0.0092 0.00874
## Cumulative Proportion 0.83555 0.84653 0.85689 0.86680 0.87643 0.8856 0.89438
##          PC29     PC30     PC31     PC32     PC33     PC34     PC35
## Standard deviation  0.67646 0.66506 0.65618 0.60972 0.59421 0.58080 0.5691
## Proportion of Variance 0.00847 0.00819 0.00797 0.00688 0.00654 0.00625 0.0060
## Cumulative Proportion 0.90285 0.91104 0.91902 0.92590 0.93244 0.93868 0.9447
##          PC36     PC37     PC38     PC39     PC40     PC41     PC42
## Standard deviation  0.55169 0.53131 0.52073 0.51261 0.50078 0.4761 0.46783
## Proportion of Variance 0.00564 0.00523 0.00502 0.00487 0.00464 0.0042 0.00405
## Cumulative Proportion 0.95032 0.95555 0.96057 0.96543 0.97008 0.9743 0.97833
##          PC43     PC44     PC45     PC46     PC47     PC48     PC49
## Standard deviation  0.45673 0.41925 0.40815 0.36105 0.32378 0.30685 0.27925
## Proportion of Variance 0.00386 0.00326 0.00309 0.00241 0.00194 0.00174 0.00144
## Cumulative Proportion 0.98219 0.98545 0.98853 0.99095 0.99289 0.99463 0.99608
##          PC50     PC51     PC52     PC53     PC54
## Standard deviation  0.27372 0.24625 0.1942 0.15436 0.12161
## Proportion of Variance 0.00139 0.00112 0.0007 0.00044 0.00027
## Cumulative Proportion 0.99746 0.99859 0.9993 0.99973 1.00000
```

```
eigenvalues <- pca_w$sdev^2
prop_var     <- eigenvalues / sum(eigenvalues)
cum_var      <- cumsum(prop_var)

pca_table <- data.frame(
  PC = paste0("PC", seq_along(eigenvalues)),
  Eigenvalue = eigenvalues,
  Proportion_of_variance = prop_var,
  Cumulative_variance = cum_var
)

pca_table
```

##	PC	Eigenvalue	Proportion_of_variance	Cumulative_variance
## 1	PC1	13.47199921	0.2494814668	0.2494815
## 2	PC2	4.91501617	0.0910188180	0.3405003
## 3	PC3	2.76978822	0.0512923744	0.3917927
## 4	PC4	2.56726255	0.0475418991	0.4393346
## 5	PC5	2.14535004	0.0397287045	0.4790633
## 6	PC6	1.85783366	0.0344043270	0.5134676
## 7	PC7	1.64725265	0.0305046786	0.5439723
## 8	PC8	1.50748634	0.0279164138	0.5718887
## 9	PC9	1.37753841	0.0255099706	0.5973987
## 10	PC10	1.36265086	0.0252342752	0.6226329
## 11	PC11	1.24522132	0.0230596541	0.6456926
## 12	PC12	1.19078172	0.0220515133	0.6677441
## 13	PC13	1.14805364	0.0212602525	0.6890043
## 14	PC14	1.04770672	0.0194019763	0.7084063
## 15	PC15	1.01988194	0.0188867026	0.7272930
## 16	PC16	0.95469474	0.0176795322	0.7449726
## 17	PC17	0.94856815	0.0175660768	0.7625386
## 18	PC18	0.91311908	0.0169096126	0.7794482
## 19	PC19	0.88787004	0.0164420378	0.7958903
## 20	PC20	0.77458899	0.0143442406	0.8102345
## 21	PC21	0.73032180	0.0135244777	0.8237590
## 22	PC22	0.63648907	0.0117868346	0.8355458
## 23	PC23	0.59312308	0.0109837607	0.8465296
## 24	PC24	0.55964367	0.0103637717	0.8568934
## 25	PC25	0.53496044	0.0099066749	0.8668000
## 26	PC26	0.52022396	0.0096337771	0.8764338
## 27	PC27	0.49677572	0.0091995504	0.8856334
## 28	PC28	0.47213324	0.0087432082	0.8943766
## 29	PC29	0.45760456	0.0084741585	0.9028507
## 30	PC30	0.44230836	0.0081908955	0.9110416
## 31	PC31	0.43057273	0.0079735690	0.9190152
## 32	PC32	0.37175610	0.0068843722	0.9258996
## 33	PC33	0.35308557	0.0065386217	0.9324382
## 34	PC34	0.33732664	0.0062467896	0.9386850
## 35	PC35	0.32390553	0.0059982505	0.9446832
## 36	PC36	0.30436391	0.0056363687	0.9503196
## 37	PC37	0.28228542	0.0052275077	0.9555471
## 38	PC38	0.27116462	0.0050215671	0.9605687
## 39	PC39	0.26276932	0.0048660985	0.9654348
## 40	PC40	0.25077733	0.0046440247	0.9700788
## 41	PC41	0.22667565	0.0041976973	0.9742765
## 42	PC42	0.21886869	0.0040531238	0.9783296
## 43	PC43	0.20860520	0.0038630593	0.9821927
## 44	PC44	0.17577010	0.0032550018	0.9854477
## 45	PC45	0.16659021	0.0030850039	0.9885327
## 46	PC46	0.13035903	0.0024140562	0.9909467
## 47	PC47	0.10483109	0.0019413166	0.9928881
## 48	PC48	0.09415453	0.0017436024	0.9946317
## 49	PC49	0.07797950	0.0014440649	0.9960757
## 50	PC50	0.07492502	0.0013875003	0.9974632
## 51	PC51	0.06063661	0.0011229001	0.9985861
## 52	PC52	0.03773256	0.0006987512	0.9992849
## 53	PC53	0.02382795	0.0004412584	0.9997261
## 54	PC54	0.01478834	0.0002738581	1.0000000

Define fertility vs. infertility using only the reproductive questionnaire answer

For comparing chemical levels between case and control groups. Two questions from reproductive questionnaire were considered as the main indicators of fertility issues. 1 - YES, 2 - NO, Teoreticky v tom popisu na jejich webovkach nabizi i hodnotny pro I dont know a refused, ale z toho popisu co tma maji je nikdo nezvolil

```
# Infertility issue variables : RHQ074 Tried for a year to become pregnant?, RHQ076 Seen  
a DR b/c unable to become pregnant?  
#survey weighted correlation for binary variables -> Phi coefficient  
x <- "RHQ074"  
y <- "RHQ076"  
f <- as.formula(paste("~", paste(c(x, y), collapse = " + ")))  
vcov_mat <- svyvar(f, obj_cleaned, na.rm = TRUE)  
sd_vec <- sqrt(diag(vcov_mat))  
phi_w <- vcov_mat[1,2] / (sd_vec[1] * sd_vec[2])  
phi_w
```

```
## RHQ074  
## 0.6701052
```

```
#number of answers that are not nan  
sum(!is.na(obj_cleaned$variables$RHQ074))
```

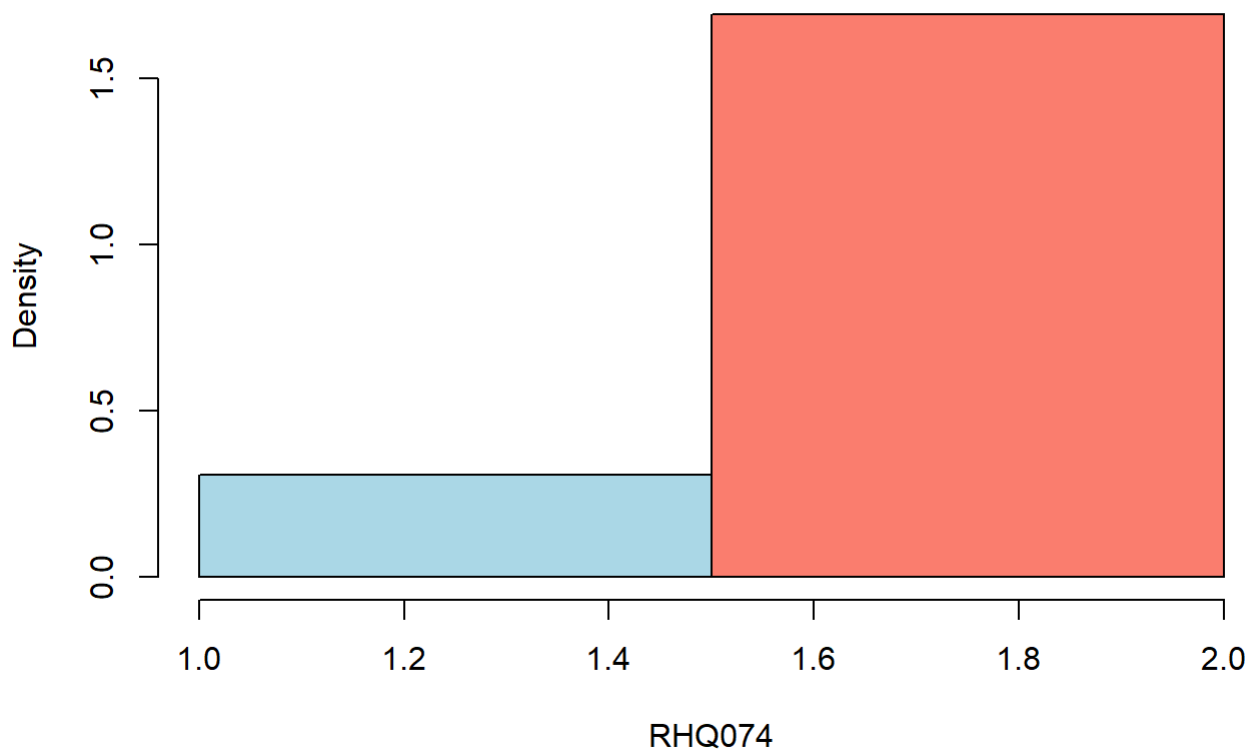
```
## [1] 286
```

```
sum(!is.na(obj_cleaned$variables$RHQ076))
```

```
## [1] 286
```

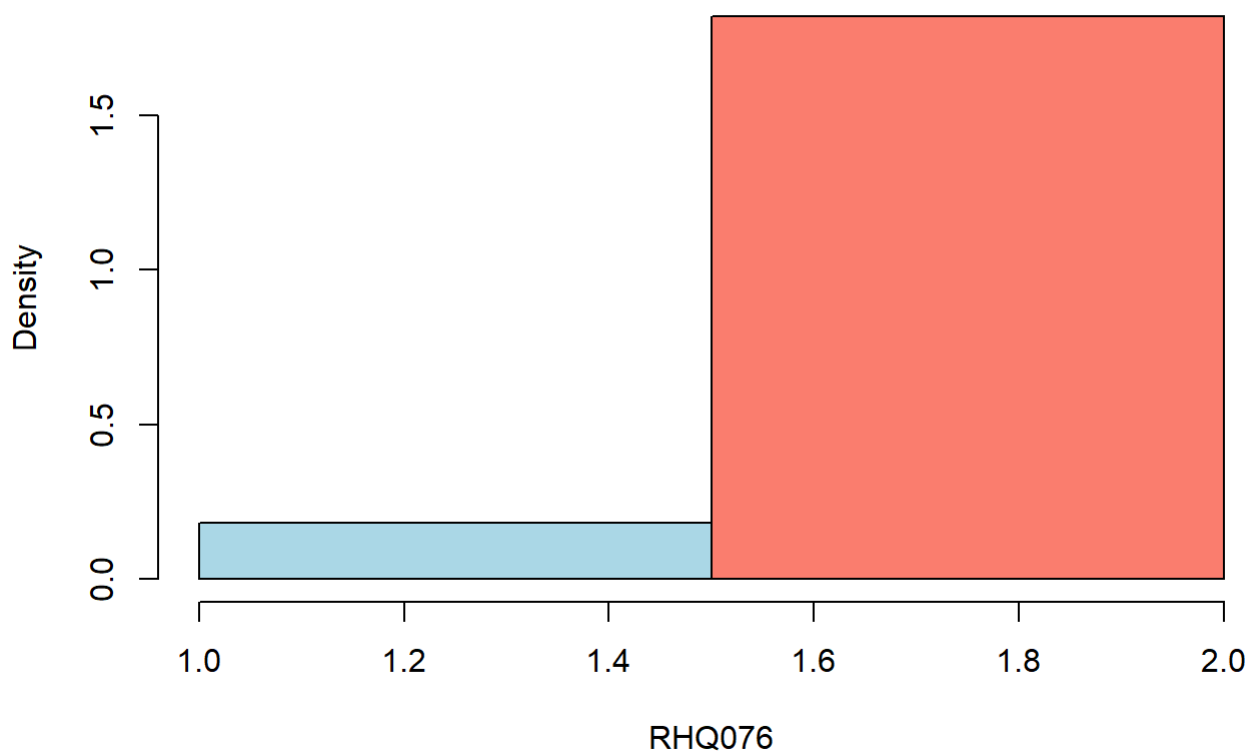
```
svyhist(~RHQ074, design = obj_cleaned, breaks = 2, col = c("lightblue", "salmon"), main = "Tried  
for a year to become pregnant? (1 = Yes, 2 = No)")
```

Tried for a year to become pregnant? (1 = Yes, 2 = No)



```
svyhist(~RHQ076, design = obj_cleaned, breaks = 2, col = c("lightblue", "salmon"), main = "Seen doctor, unable to become pregnant ? (1 = Yes, 2 = No)")
```

Seen doctor, unable to become pregnant ? (1 = Yes, 2 = No)



```
#Kolik radku melo odpoved ano (Na to kolik to reprezentuje lidi by jsme asi museli pocitat s
tema vahama)
sum(obj_cleaned$variables$RHQ074 == 1, na.rm = TRUE)
```

```
## [1] 37
```

Chemical concentrations in case/control groups

Case group (infertile) lide kteri odpovedeli ano na Tried for 12 months to become pregnant. Control group (fertile) odpoveded Ne The selection of chemicals presented in the boxplots below was based on previously published studies (references, ty co tam uz jsou v planu), which frequently focused on Bisphenol A, Triclosan, and MEHP among other compounds. The fourth phthalate choice was based on online resources.

z clanku (Screening for phthalates biomarkers and its potential role in infertility outcomes in Jordan) The graphs shows boxplots for four different compounds. The middle line represents the median while upper and lower box borders represent the first and third quartiles. The whiskers represent minimum and maximum value. The p-value below each graph is from Wilcoxon non parametric test. P-values less than 0.005 were considered significant. The test was performed using survey weighted Wilcoxon rank-sum test -> correct approach to account for NHANES complex sampling design.

```
#URXBPH - bisphenol A
table(obj_cleaned$variables$RHQ074, useNA = "ifany")
```

```
##
##      1      2 <NA>
##    37   249    38
```

```
summary(obj_cleaned$variables$log_URXBPH)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.1310  0.5306  0.8542  0.9759  1.3350  4.1495
```

```

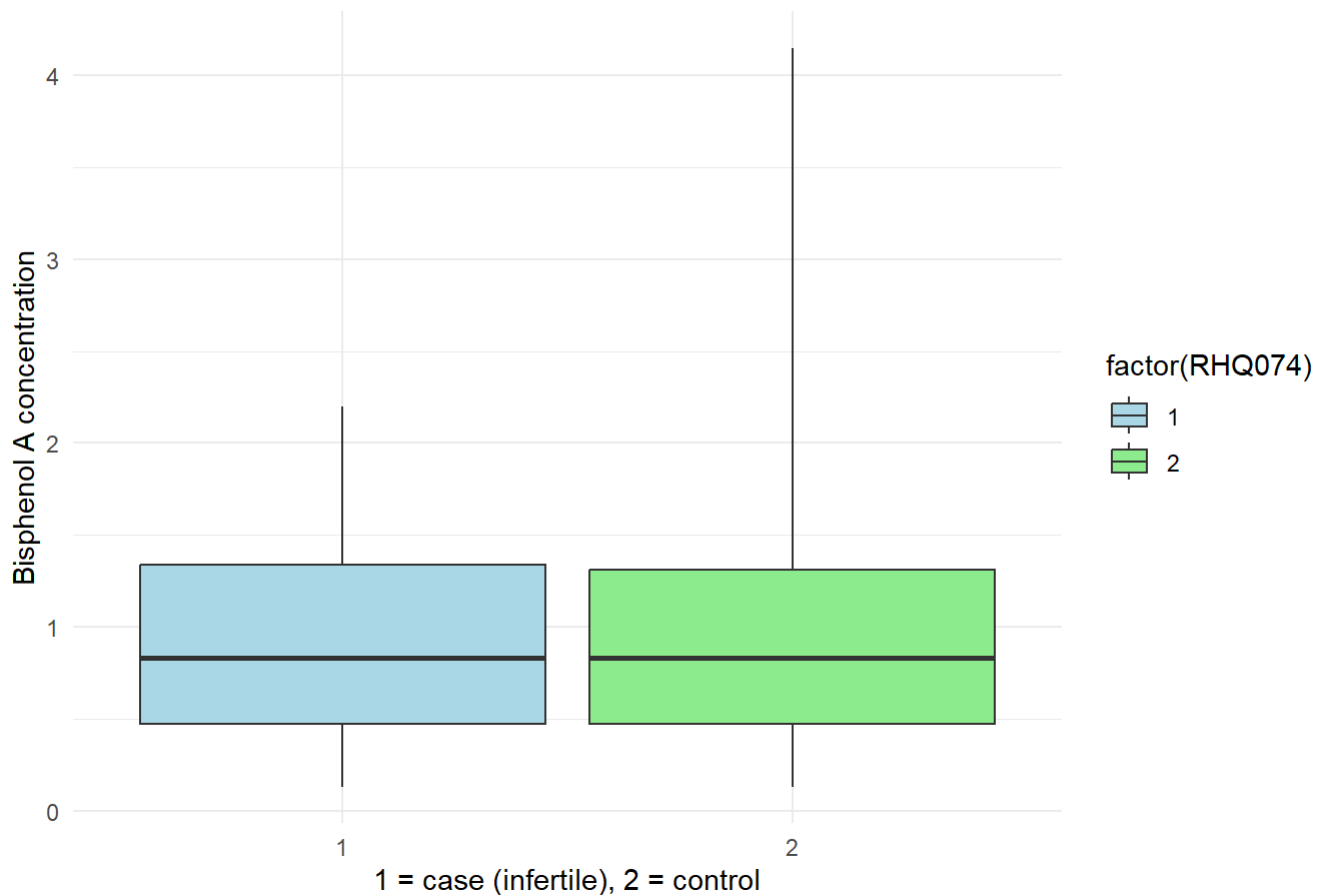
subset_design <- subset(obj_cleaned, !is.na(RHQ074) & !is.na(log_URXBPH))

design_by_group <- svyby(
  ~log_URXBPH,
  ~RHQ074,
  subset_design,
  FUN = svyquantile,
  quantiles = c(0, 0.25, 0.5, 0.75, 1),
  na.rm = TRUE,
  ci = FALSE,          # Turn off confidence intervals first
  keep.var = FALSE
)
df_box <- design_by_group
names(df_box) <- c("RHQ074", "min", "Q1", "median", "Q3", "max")
df_box <- df_box %>% select(RHQ074, min, Q1, median, Q3, max)

ggplot(df_box, aes(x = factor(RHQ074), fill = factor(RHQ074))) +
  geom_boxplot(
    aes(
      ymin = min,
      lower = Q1,
      middle = median,
      upper = Q3,
      ymax = max
    ),
    stat = "identity"
  ) +
  scale_fill_manual(values = c("lightblue", "lightgreen")) +
  ylab("Bisphenol A concentration") +
  xlab("1 = case (infertile), 2 = control") +
  ggtitle("Weighted boxplot for case and control difference in log transformed Bisphenol A co
ncentration") +
  theme_minimal()

```

Weighted boxplot for case and control difference in log transformed Bisphenol A con



```
#Wilcoxon test
subset_design <- subset(
  obj_cleaned,
  !is.na(log_URXBPH) & !is.na(RHQ074)
)

wt <- svyranktest(
  log_URXBPH ~ factor(RHQ074),
  design = subset_design,
  test = "wilcoxon"
)
cat("Wilcoxon p-value:", signif(wt$p.value, 3), "\n")
```

```
## Wilcoxon p-value: 0.696
```

```
#URXTRS - triclosan
table(obj_cleaned$variables$RHQ074, useNA = "ifany")
```

```
##
##    1    2 <NA>
##   37  249   38
```

```
summary(obj_cleaned$variables$log_URXTRS)
```

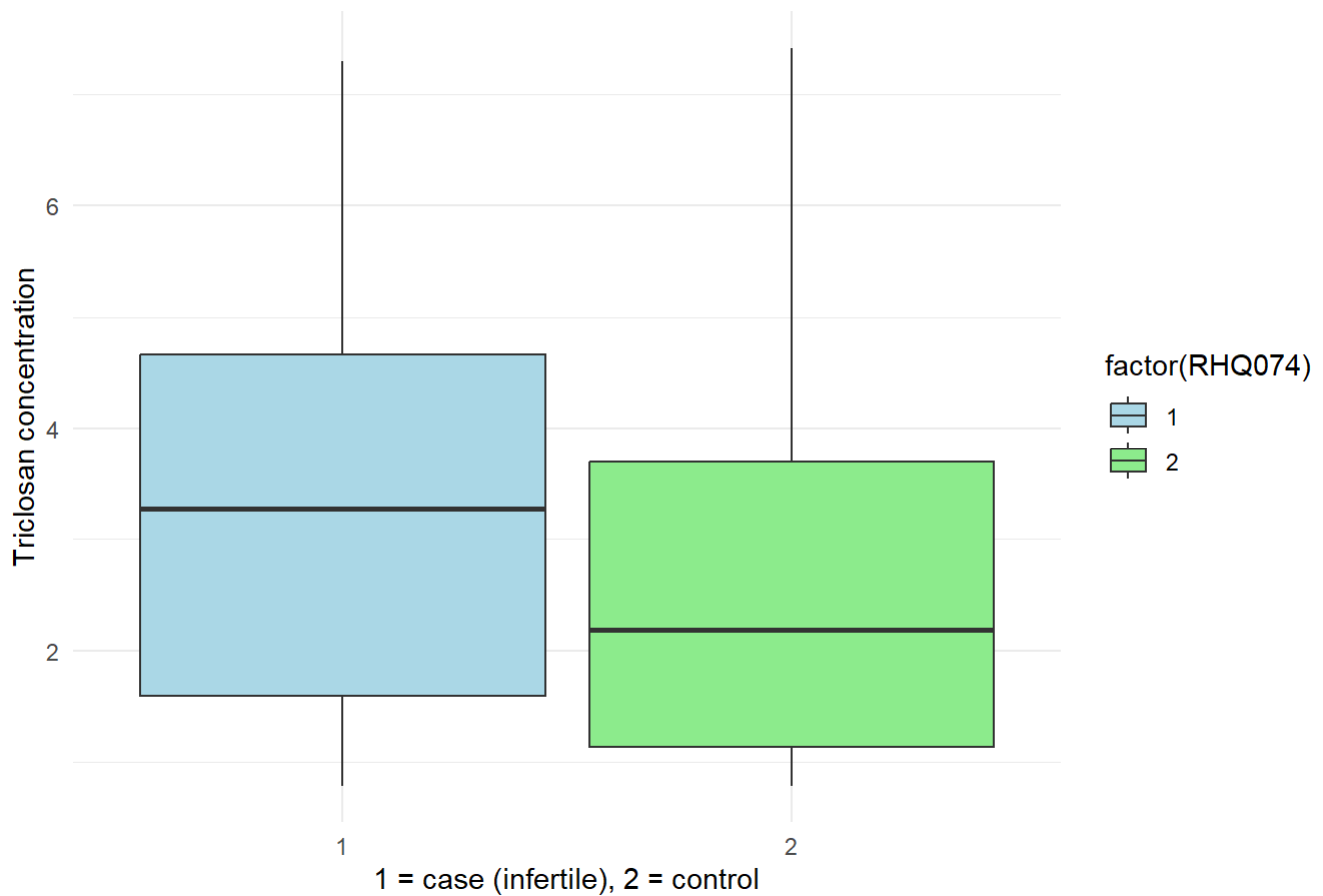
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.7885  1.1862  2.1861  2.6402  3.7075  7.4152
```

```
subset_design <- subset(obj_cleaned, !is.na(RHQ074) & !is.na(log_URXTRS))

design_by_group <- svyby(
  ~log_URXTRS,
  ~RHQ074,
  subset_design,
  FUN = svyquantile,
  quantiles = c(0, 0.25, 0.5, 0.75, 1),
  na.rm = TRUE,
  ci = FALSE,          # Turn off confidence intervals first
  keep.var = FALSE
)
df_box <- design_by_group
names(df_box) <- c("RHQ074", "min", "Q1", "median", "Q3", "max")
df_box <- df_box %>% select(RHQ074, min, Q1, median, Q3, max)

ggplot(df_box, aes(x = factor(RHQ074), fill = factor(RHQ074))) +
  geom_boxplot(
    aes(
      ymin = min,
      lower = Q1,
      middle = median,
      upper = Q3,
      ymax = max
    ),
    stat = "identity"
  ) +
  scale_fill_manual(values = c("lightblue", "lightgreen")) +
  ylab("Triclosan concentration") +
  xlab("1 = case (infertile), 2 = control") +
  ggtitle("Weighted boxplot for case and control difference in log transformed Triclosan concentration") +
  theme_minimal()
```

Weighted boxplot for case and control difference in log transformed Triclosan conce



```
#Wilcoxon test
subset_design <- subset(
  obj_cleaned,
  !is.na(log_URXTRS) & !is.na(RHQ074)
)

wt <- svyranktest(
  log_URXTRS ~ factor(RHQ074),
  design = subset_design,
  test = "wilcoxon"
)
cat("Wilcoxon p-value:", signif(wt$p.value, 3), "\n")
```

```
## Wilcoxon p-value: 0.0364
```

```
#URXMHH - MEHP
table(obj_cleaned$variables$RHQ074, useNA = "ifany")
```

```
##
##    1    2 <NA>
##   37  249   38
```

```
summary(obj_cleaned$variables$log_URXMHH)
```

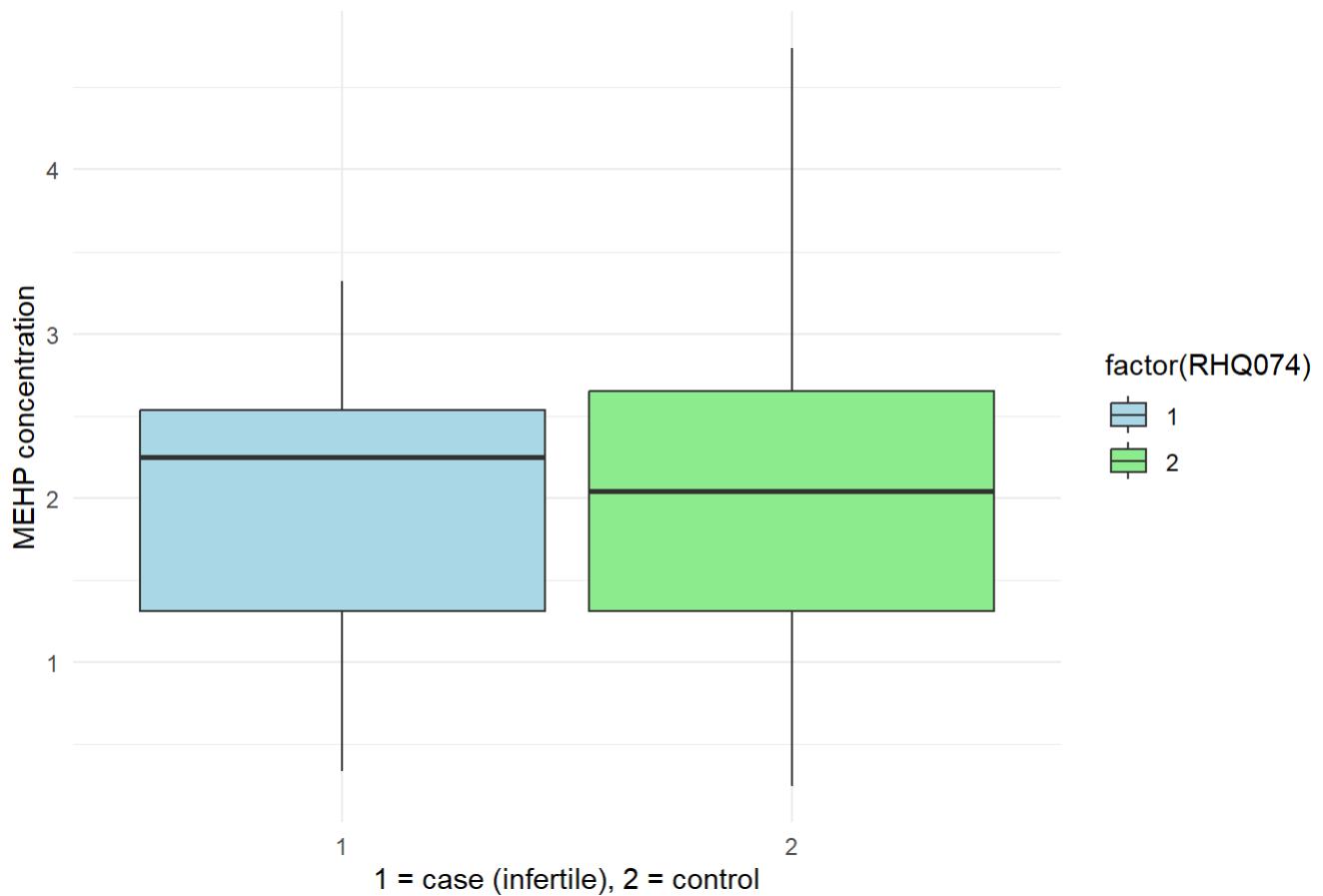
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2469  1.3800  2.0669  2.0318  2.6479  4.7423
```

```
subset_design <- subset(obj_cleaned, !is.na(RHQ074) & !is.na(log_URXMHH))

design_by_group <- svyby(
  ~log_URXMHH,
  ~RHQ074,
  subset_design,
  FUN = svyquantile,
  quantiles = c(0, 0.25, 0.5, 0.75, 1),
  na.rm = TRUE,
  ci = FALSE,          # Turn off confidence intervals first
  keep.var = FALSE
)
df_box <- design_by_group
names(df_box) <- c("RHQ074", "min", "Q1", "median", "Q3", "max")
df_box <- df_box %>% select(RHQ074, min, Q1, median, Q3, max)

ggplot(df_box, aes(x = factor(RHQ074), fill = factor(RHQ074))) +
  geom_boxplot(
    aes(
      ymin = min,
      lower = Q1,
      middle = median,
      upper = Q3,
      ymax = max
    ),
    stat = "identity"
  ) +
  scale_fill_manual(values = c("lightblue", "lightgreen")) +
  ylab("MEHP concentration") +
  xlab("1 = case (infertile), 2 = control") +
  ggtitle("Weighted boxplot for case and control difference in log transformed MEHP concentra
tion") +
  theme_minimal()
```

Weighted boxplot for case and control difference in log transformed MEHP concentr



```
#Wilcoxon test
subset_design <- subset(
  obj_cleaned,
  !is.na(log_URXMHH) & !is.na(RHQ074)
)

wt <- svyranktest(
  log_URXMHH ~ factor(RHQ074),
  design = subset_design,
  test = "wilcoxon"
)
cat("Wilcoxon p-value:", signif(wt$p.value, 3), "\n")
```

```
## Wilcoxon p-value: 0.84
```

```
#URXMPB - monobutyl phtalate
table(obj_cleaned$variables$RHQ074, useNA = "ifany")
```

```
##
##      1      2 <NA>
##    37   249    38
```

```
summary(obj_cleaned$variables$log_URXMPB)
```

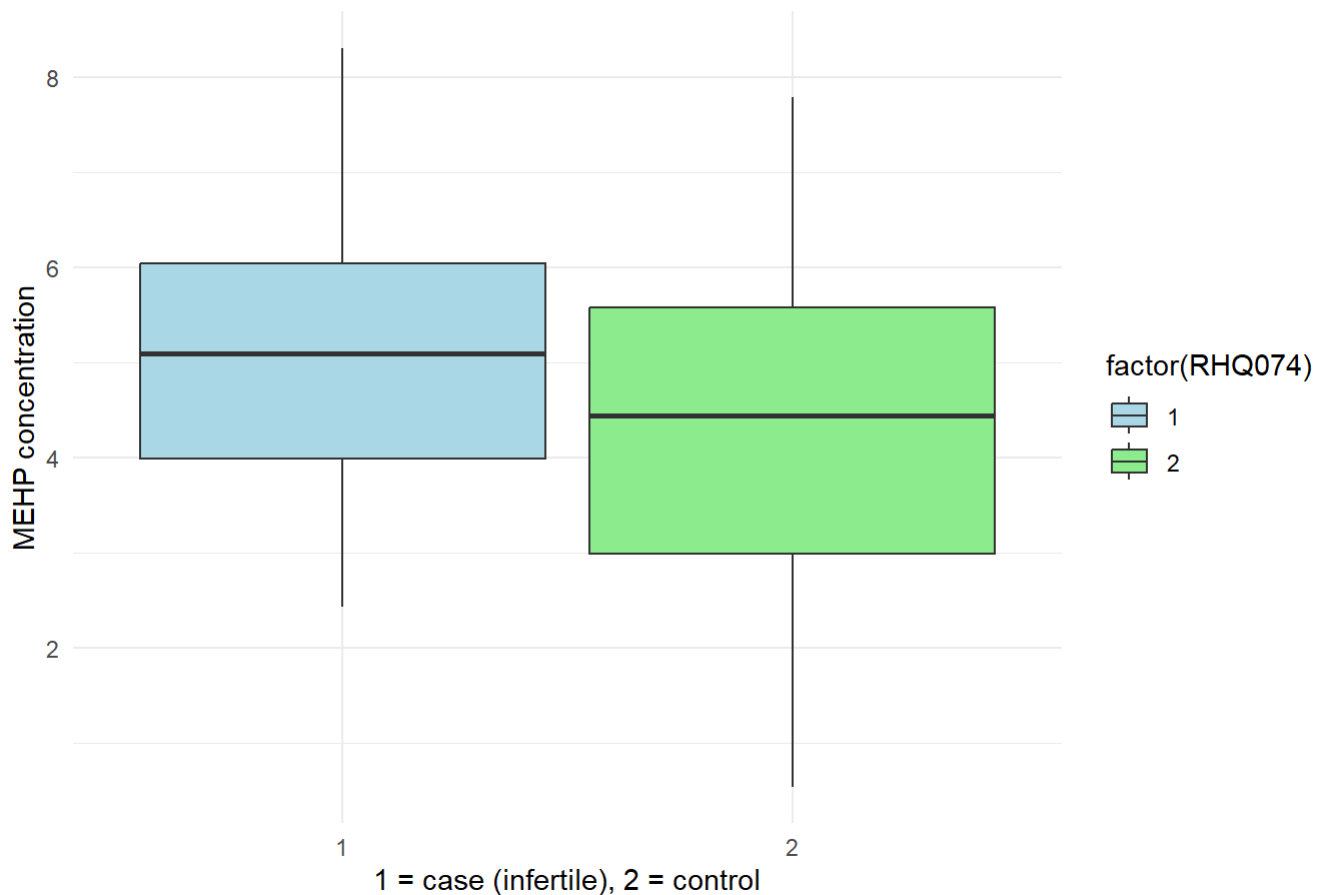
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5365  3.3810  4.7829  4.5640  5.6931  8.3073
```

```
subset_design <- subset(obj_cleaned, !is.na(RHQ074) & !is.na(log_URXMPB))

design_by_group <- svyby(
  ~log_URXMPB,
  ~RHQ074,
  subset_design,
  FUN = svyquantile,
  quantiles = c(0, 0.25, 0.5, 0.75, 1),
  na.rm = TRUE,
  ci = FALSE,          # Turn off confidence intervals first
  keep.var = FALSE
)
df_box <- design_by_group
names(df_box) <- c("RHQ074", "min", "Q1", "median", "Q3", "max")
df_box <- df_box %>% select(RHQ074, min, Q1, median, Q3, max)

ggplot(df_box, aes(x = factor(RHQ074), fill = factor(RHQ074))) +
  geom_boxplot(
    aes(
      ymin = min,
      lower = Q1,
      middle = median,
      upper = Q3,
      ymax = max
    ),
    stat = "identity"
  ) +
  scale_fill_manual(values = c("lightblue", "lightgreen")) +
  ylab("MEHP concentration") +
  xlab("1 = case (infertile), 2 = control") +
  ggtitle("Weighted boxplot for case and control difference in log transformed MPB concentration") +
  theme_minimal()
```

Weighted boxplot for case and control difference in log transformed MPB concentrat



```
#Wilcoxon test
subset_design <- subset(
  obj_cleaned,
  !is.na(log_URXMPB) & !is.na(RHQ074)
)

wt <- svyranktest(
  log_URXMPB ~ factor(RHQ074),
  design = subset_design,
  test = "wilcoxon"
)
cat("Wilcoxon p-value:", signif(wt$p.value, 3), "\n")
```

```
## Wilcoxon p-value: 0.0536
```

Compare fertility vs. infertility with hormone levels

```
table(obj_cleaned$variables$RHQ074, useNA = "ifany")
```

```
##
##    1    2 <NA>
##   37  249   38
```

```
summary(obj_cleaned$variables$log_LBXEST)
```

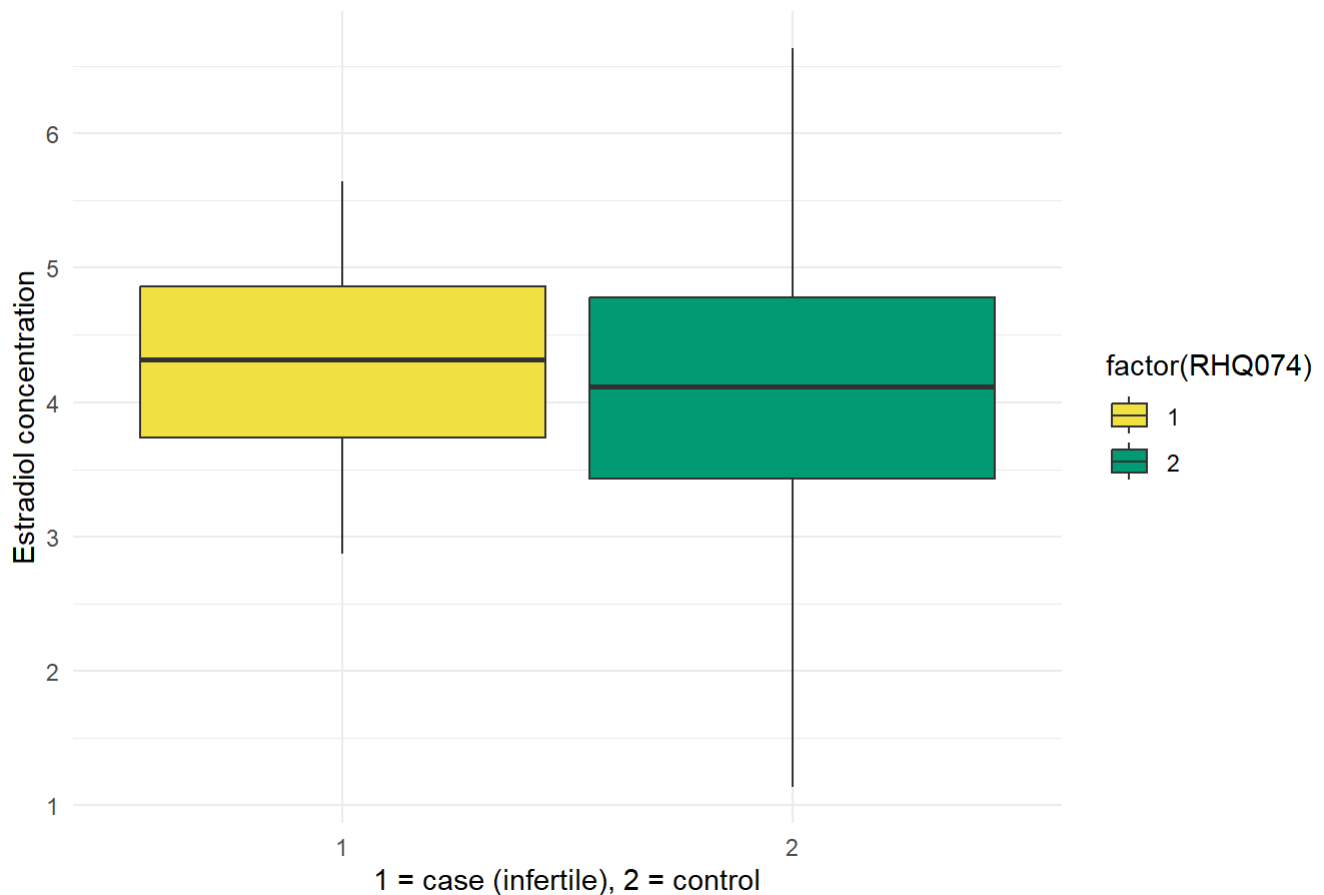
##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
##	1.137	3.597	4.274	4.133	4.852	6.639	5

```
subset_design <- subset(obj_cleaned, !is.na(RHQ074) & !is.na(log_LBXEST))

design_by_group <- svyby(
  ~log_LBXEST,
  ~RHQ074,
  subset_design,
  FUN = svyquantile,
  quantiles = c(0, 0.25, 0.5, 0.75, 1),
  na.rm = TRUE,
  ci = FALSE,          # Turn off confidence intervals first
  keep.var = FALSE
)
df_box <- design_by_group
names(df_box) <- c("RHQ074", "min", "Q1", "median", "Q3", "max")
df_box <- df_box %>% select(RHQ074, min, Q1, median, Q3, max)

ggplot(df_box, aes(x = factor(RHQ074), fill = factor(RHQ074))) +
  geom_boxplot(
    aes(
      ymin = min,
      lower = Q1,
      middle = median,
      upper = Q3,
      ymax = max
    ),
    stat = "identity"
  ) +
  scale_fill_manual(values = c("#F0E442", "#009E73")) +
  ylab("Estradiol concentration") +
  xlab("1 = case (infertile), 2 = control") +
  ggtitle("Weighted boxplot for case and control difference in log transformed Estradiol hormone concentration") +
  theme_minimal()
```

Weighted boxplot for case and control difference in log transformed Estradiol hormone



```
#Wilcoxon test
subset_design <- subset(
  obj_cleaned,
  !is.na(log_LBXEST) & !is.na(RHQ074)
)

wt <- svyranktest(
  log_LBXEST ~ factor(RHQ074),
  design = subset_design,
  test = "wilcoxon"
)
cat("Wilcoxon p-value:", signif(wt$p.value, 3), "\n")
```

```
## Wilcoxon p-value: 0.27
```

```
table(obj_cleaned$variables$RHQ074, useNA = "ifany")
```

```
##
##    1    2 <NA>
##   37  249   38
```

```
summary(obj_cleaned$variables$log_LBXSHBG)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##    2.423  3.724   4.109   4.129  4.460   5.710    41
```

```

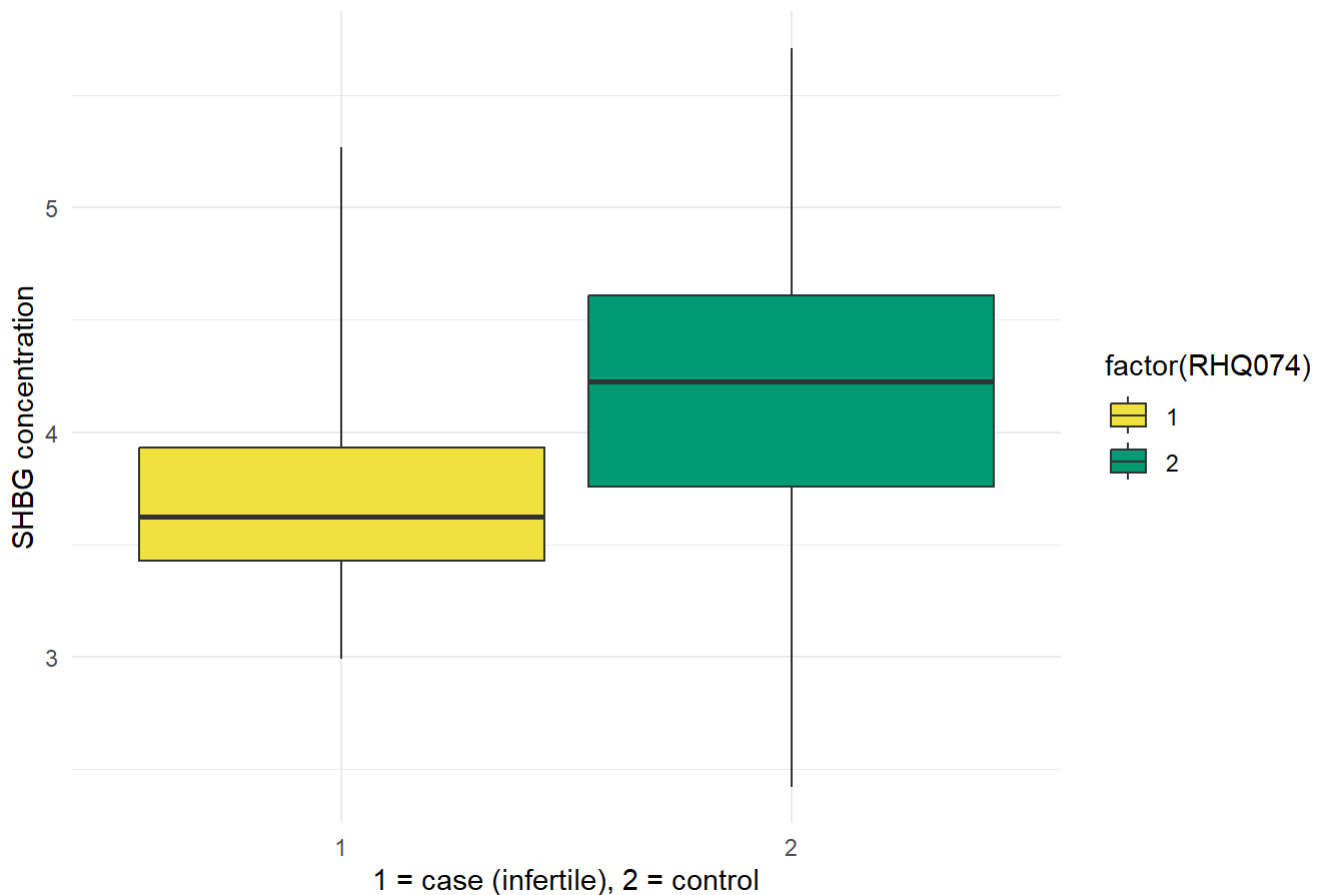
subset_design <- subset(obj_cleaned, !is.na(RHQ074) & !is.na(log_LBXSHBG))

design_by_group <- svyby(
  ~log_LBXSHBG,
  ~RHQ074,
  subset_design,
  FUN = svyquantile,
  quantiles = c(0, 0.25, 0.5, 0.75, 1),
  na.rm = TRUE,
  ci = FALSE,          # Turn off confidence intervals first
  keep.var = FALSE
)
df_box <- design_by_group
names(df_box) <- c("RHQ074", "min", "Q1", "median", "Q3", "max")
df_box <- df_box %>% select(RHQ074, min, Q1, median, Q3, max)

ggplot(df_box, aes(x = factor(RHQ074), fill = factor(RHQ074))) +
  geom_boxplot(
    aes(
      ymin = min,
      lower = Q1,
      middle = median,
      upper = Q3,
      ymax = max
    ),
    stat = "identity"
  ) +
  scale_fill_manual(values = c("#F0E442", "#009E73")) +
  ylab("SHBG concentration") +
  xlab("1 = case (infertile), 2 = control") +
  ggtitle("Weighted boxplot for case and control difference in log transformed SHBG hormone c
oncentration") +
  theme_minimal()

```

Weighted boxplot for case and control difference in log transformed SHBG hormone



```
#Wilcoxon test
subset_design <- subset(
  obj_cleaned,
  !is.na(log_LBXSHBG) & !is.na(RHQ074)
)

wt <- svyranktest(
  log_LBXSHBG ~ factor(RHQ074),
  design = subset_design,
  test = "wilcoxon"
)
cat("Wilcoxon p-value:", signif(wt$p.value, 3), "\n")
```

```
## Wilcoxon p-value: 0.0142
```

```
table(obj_cleaned$variables$RHQ074, useNA = "ifany")
```

```
##
##    1    2 <NA>
##   37  249   38
```

```
summary(obj_cleaned$variables$log_LBXTST)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.376  2.861   3.178   3.173  3.464   6.356
```

```

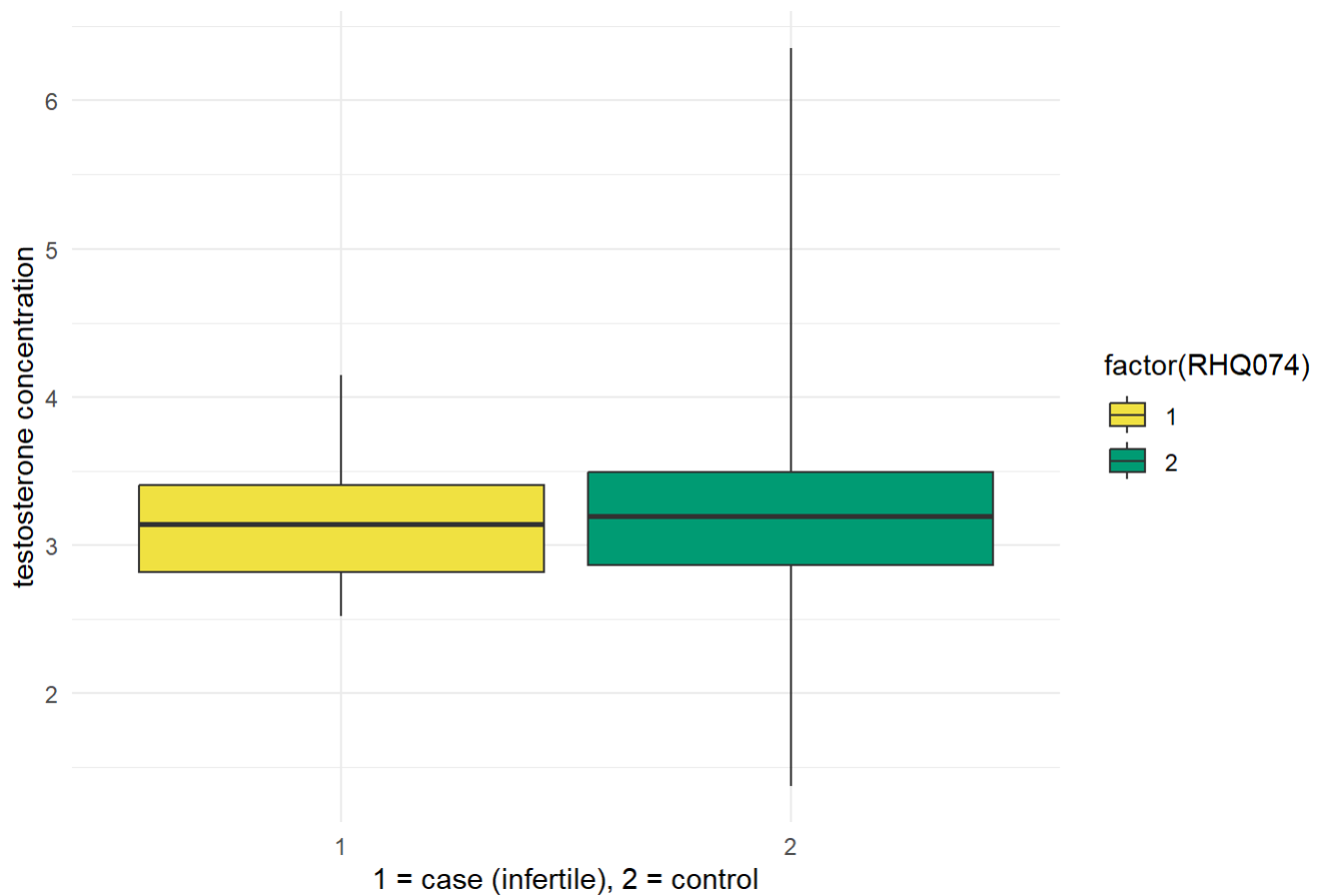
subset_design <- subset(obj_cleaned, !is.na(RHQ074) & !is.na(log_LBXTST))

design_by_group <- svyby(
  ~log_LBXTST,
  ~RHQ074,
  subset_design,
  FUN = svyquantile,
  quantiles = c(0, 0.25, 0.5, 0.75, 1),
  na.rm = TRUE,
  ci = FALSE,          # Turn off confidence intervals first
  keep.var = FALSE
)
df_box <- design_by_group
names(df_box) <- c("RHQ074", "min", "Q1", "median", "Q3", "max")
df_box <- df_box %>% select(RHQ074, min, Q1, median, Q3, max)

ggplot(df_box, aes(x = factor(RHQ074), fill = factor(RHQ074))) +
  geom_boxplot(
    aes(
      ymin = min,
      lower = Q1,
      middle = median,
      upper = Q3,
      ymax = max
    ),
    stat = "identity"
  ) +
  scale_fill_manual(values = c("#F0E442", "#009E73")) +
  ylab("testosterone concentration") +
  xlab("1 = case (infertile), 2 = control") +
  ggtitle("Weighted boxplot for case and control difference in log transformed testosterone hormone concentration") +
  theme_minimal()

```

Weighted boxplot for case and control difference in log transformed testosterone hor



```
#Wilcoxon test
subset_design <- subset(
  obj_cleaned,
  !is.na(log_LBXTST) & !is.na(RHQ074)
)

wt <- svyranktest(
  log_LBXTST ~ factor(RHQ074),
  design = subset_design,
  test = "wilcoxon"
)
cat("Wilcoxon p-value:", signif(wt$p.value, 3), "\n")
```

```
## Wilcoxon p-value: 0.348
```

```
colnames(obj_cleaned)
```

##	[1]	"SEQN"	"RIDAGEYR"	"RIDEXPRG"	"DMDEDUC2"	"RIAGENDR"
##	[6]	"INDFMPIR"	"WTMEC2YR"	"SDMVPSU"	"SDMVSTRA"	"RHQ031"
##	[11]	"RHQ074"	"RHQ076"	"RHQ131"	"RHQ160"	"RHQ171"
##	[16]	"RHD280"	"RHQ078"	"RHD180"	"LBXTST"	"LBDTSTLC"
##	[21]	"LBXEST"	"LBDESTLC"	"LBXSHBG"	"LBDSHGLC"	"WTSSBH2Y"
##	[26]	"URXUCR"	"SSURMHBP"	"SDUMHBPL"	"SSURHIBP"	"SDUHIBPL"
##	[31]	"WTSB2YR"	"URXBP3"	"URDBP3LC"	"URXBPH"	"URDBPHLC"
##	[36]	"URXBPF"	"URDBPFLC"	"URXBPS"	"URDBPSLC"	"URXTLC"
##	[41]	"URDTLCLC"	"URXTRS"	"URDTRSLC"	"URXBUP"	"URDBUPLC"
##	[46]	"URXEPB"	"URDEPBLC"	"URXMPB"	"URDMPBLC"	"URXPPB"
##	[51]	"URDPPBLC"	"URX14D"	"URD14DLC"	"URXDCB"	"URDDCBLC"
##	[56]	"ALQ120Q"	"SMQ040"	"SMD650"	"BMXBMI"	"URXUCR.1"
##	[61]	"URXCNP"	"URDCNPLC"	"URXCOP"	"URDCOPLC"	"URXECF"
##	[66]	"URDECPLC"	"URXMBP"	"URDMBPLC"	"URXMC1"	"URDMC1LC"
##	[71]	"URXMEP"	"URDMEPLC"	"URXMHH"	"URDMHHLC"	"URXMHNC"
##	[76]	"URDMCHLC"	"URXMHP"	"URDMHPLC"	"URXMIB"	"URDMIBLC"
##	[81]	"URXMNP"	"URDMNPLC"	"URXMOH"	"URDMOHLIC"	"URXMZP"
##	[86]	"URDMZPLC"	"log_URXUCR"	"log_SSURMHBP"	"log_SDUMHBPL"	"log_SSURHIBP"
##	[91]	"log_SDUHIBPL"	"log_URXBP3"	"log_URDBP3LC"	"log_URXBPH"	"log_URDBPHLC"
##	[96]	"log_URXBPF"	"log_URDBPFLC"	"log_URXBPS"	"log_URDBPSLC"	"log_URXTLC"
##	[101]	"log_URDTLCLC"	"log_URXTRS"	"log_URDTRSLC"	"log_URXBUP"	"log_URDBUPLC"
##	[106]	"log_URXEPB"	"log_URDEPBLC"	"log_URXMPB"	"log_URDMPBLC"	"log_URXPPB"
##	[111]	"log_URDPPBLC"	"log_URX14D"	"log_URD14DLC"	"log_URXDCB"	"log_URDDCBLC"
##	[116]	"log_URXCNP"	"log_URDCNPLC"	"log_URXCOP"	"log_URDCOPLC"	"log_URXECF"
##	[121]	"log_URDECPLC"	"log_URXMBP"	"log_URDMBPLC"	"log_URXMC1"	"log_URDMC1LC"
##	[126]	"log_URXMEP"	"log_URDMEPLC"	"log_URXMHH"	"log_URDMHHLC"	"log_URXMHNC"
##	[131]	"log_URDMCHLC"	"log_URXMHP"	"log_URDMHPLC"	"log_URXMIB"	"log_URDMIBLC"
##	[136]	"log_URXMNP"	"log_URDMNPLC"	"log_URXMOH"	"log_URDMOHLIC"	"log_URXMZP"
##	[141]	"log_URDMZPLC"	"log_LBXEST"	"log_LBXSHBG"	"log_LBXTST"	